Unit 3: Renewable Energy

Reading: Emails about renewable energy

Lesson 3-1
Book pages 25-27

💡 **Big idea:** **Renewable energy** comes from sources that never **run out**, like the sun, wind and water. We read **emails** between Luke, who wants to invent a trolley powered by renewable energy, and his uncle Bill, who sends information about **types of renewable energy**.

🎯 **I will learn:**

- Sort renewable and non-renewable energy sources
- Read emails and identify their purpose
- Learn the types of renewable energy
- Suggest solutions to problems

📖 Key words

renewable متجدد	**non-renewable** غير متجدد	**run out** ينفد	**solar power** الطاقة الشمسية
wind power طاقة الرياح	**hydropower** الطاقة الكهرومائية	**geothermal** حرارية جوفية	**biofuel** وقود حيوي
turbine توربين	**generate** يولّد	**convert** يحوّل	**invention** اختراع

Renewable or non-renewable?

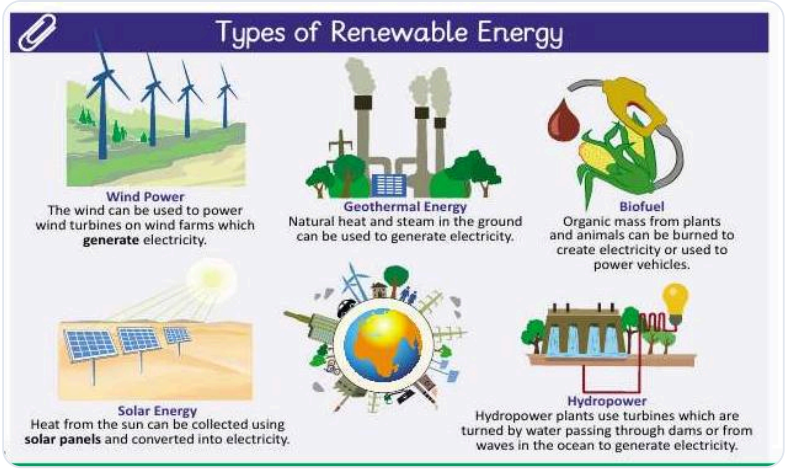
- **Renewable**: wind, the sun, biomass, water, geothermal heat
- **Non-renewable**: oil, coal, natural gas, nuclear power
- Renewable energy is **better for the environment**
- Non-renewable energy will **run out**

Luke's email

- Purpose: to **ask for help** with a school project
- Problem: the shopping trolley is heavy and hard to push
- Idea: an **automatic trolley** powered by renewable energy
- A small panel converts light from the mall into electricity

Types of renewable energy

- **Wind power**: wind turbines on wind farms generate electricity
- **Geothermal**: natural heat and steam in the ground
- **Biofuel**: organic mass from plants and animals
- **Solar**: heat from the sun collected by solar panels
- **Hydropower**: water through dams turns turbines



Types of renewable energy

✏️ Examples

Is natural gas renewable?

Answer: No, it is non-renewable because it will run out.

What is the purpose of Luke's email?

Answer: To ask Bill for help with his school project.

Complete: ___ are used on wind farms and hydropower plants.

Answer: Turbines

⭐ Remember

- Renewable = comes back again (re-new)
- Read the subject line of an email to guess its purpose

⚠️ Watch out

- ✗ Solar energy will run out. ✓ Solar energy is renewable
- ✗ Biofuel is produced using water. ✓ Biofuel comes from organic mass

🧩 **Activity:** Sort ten energy sources into renewable and non-renewable

✅ Check yourself

1. Name three renewable sources
2. What converts sunlight into electricity?
3. Meaning of generate?

Answers: (1) sun, wind, water ، (2) solar panels ، (3) produce